LESSON 4.8
ABSOLUTE VALUE – PART 1

LESSON INTRODUCTION

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

LEARNING TARGETS

- I can find the absolute value of a rational number.

BENCHMARK COHERENCE

WORKING TOWARD

MA.7.NSO.2.1 Solve mathematical problems using multi-step order of operations with rational numbers including grouping symbols, whole-number exponents and absolute value.

ESSENTIAL QUESTIONS

- How can you determine any number's distance from zero on a number line?
- How do you determine the absolute value of a number?

LESSON NARRATIVE

Students formally define and make meaning of absolute value in this first of a two-part lesson series introducing absolute value. Both Lessons 8 and 9 target MA.6.NSO.1.3, with Lesson 8 highlighting the formal definition of absolute value as students practice identifying it and Lesson 9 extending that learning to include making meaning of absolute values in real-world contexts.

COMPONENT SUMMARY

4.8.1 WARM-UP *(~5 MIN.)*

Justify which stack of coins they would rather have

4.8.2 EXPLORATION *(5-10 MIN.)*

Exploration of distance to zero and opposites using a number line

4.8.3 GUIDED INSTRUCTION *(5 MIN.)*

Learn the meaning of absolute value

4.8.4 GUIDED PRACTICE *(10 MIN.)*

Guided practice determining the absolute value using a number line

4.8.5 YOUR TURN! *(10 MIN.)*

Additional practice determining the absolute value using a number line

4.8.6 WRAP-UP *(~5 MIN.)*

Self-assessment of mastery of lesson learning target

RESOURCES

GLOSSARY

Key Terms:

- Absolute value

Additional Terms:

- N/A

MATERIALS

- Number line templates

ADDITIONAL PREPARATION

- 4.8.2 Exploration prompts students to use a bug cutout in the number line example.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle estimating the location of fractions and decimals when plotting rational numbers on the number line.
- Students may incorrectly think that the distance of a negative integer from zero is a negative number or that the absolute value of a number is its opposite.

THINGS TO CONSIDER

- Avoid defining absolute value as the positive value, as this does not build conceptual understanding, which will have impacts for their understanding of adding and subtracting integers in Unit 7. Focus instead on students making sense of absolute value as the number's distance from zero.
- Students may benefit from having number lines to reference throughout the lesson.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 4.8.3 Guided Instruction, ask students what they notice and wonder about the absolute values of opposite numbers. Facilitate a discussion to ensure students understand why the absolute values of opposites are equal.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Error Analysis

4.8.4 Guided Practice provides an opportunity for students to discuss student work. Students discuss how it could be possible for more than one student answer to be correct to a single problem. Students will build a stronger understanding of the concepts through this discussion.

DIFFERENTIATE INSTRUCTION

4.8.2 Exploration provides kinesthetic entry points for students who better make sense of concepts using manipulatives and real-world spatial awareness. Consider differentiated approaches to this activity based on your students.

- Utilize the bug cutout and number line for students to move along the line.
- Construct a life-size number line with students acting as the bugs.

4.8.1 WARM-UP: WOULD YOU RATHER?

Component Overview: *Students determine which stack of coins they would prefer. Then students justify their selection.*

TEACHER MOVES

Contemplate, then Calculate

Use this instructional routine to position students as thinkers first, thinking about their approach before immediately jumping into the calculations. This routine provides space and opportunity for metacognition, or thinking about their thinking, which ultimately contributes toward fluency.

4.8.1 Warm-Up: Would You Rather?

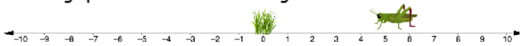
1. Would you rather have Pile A or Pile B? Justify your reasoning.

Pile A	Pile B
14 pennies	2 pennies
7 nickels	4 nickels
2 dimes	2 dimes
1 quarter	2 quarters

Answers will vary. I would rather have Pile A because Pile A has $0.94 and Pile B has $0.92.

4.8.2 Exploration: Jumping Bugs

A bug is jumping around on the number line. His home is located at 0. Use a cut out of the bug to answer the following questions about the bug's location.



1. The bug starts at 5 and jumps 3 units to the right.
   - a. What is the bug's new location?
     8
   - b. How far away from home is the bug?
     8 *jumps*
2.
   - a. If the bug starts at 2 and jumps 5 units to the left, what is the bug's new location?
     −3
   - b. How far away from home is the bug?
     3 *jumps*
3. Mr. Rodriguez told his class to begin with the bug at home. Then the bug should jump 4 units. When Micah and Brianna did this, they discovered that their bugs were in two different locations. Discuss with your partner how this may have happened. Write your explanation or draw a picture of what could have happened.
   Answers will vary. Micah and Brianna jumped their bugs in opposite directions, landing on −4 and 4.

4.8.2 EXPLORATION: JUMPING BUGS

Component Overview: *Students explore the distance to zero of several integers, including opposites, using a number line. The exploration occurs through the story of a bug that jumps to the left and the right. The bug's home is at zero.*

QUESTIONING

Try this activity outside to better engage students in making sense of distance between units, direction of units, and more.

4.8.3 GUIDED INSTRUCTION: ABSOLUTE VALUE AND ZERO

Component Overview: *Students complete guided notes as they learn the meaning of absolute value. Consider creating and displaying an anchor chart with instructions, examples, and visual models for students to reference in the subsequent components.*

TEACHER MOVES

Notice and Wonder

Ask students what they notice and wonder about the absolute values of opposite numbers. Facilitate a discussion to ensure students understand why the absolute values of opposites are equal.

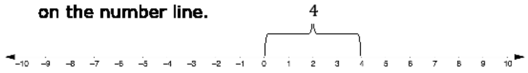
4.8.3 Guided Instruction: Absolute Value and Zero

Recall how opposite numbers relate to zero. Complete the explanation below by writing a word in each blank.

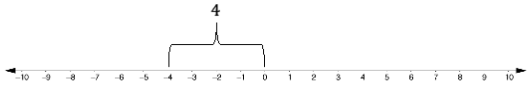
1. **Pairs of opposite values are the same *distance* from zero on the number line. They are mirror images of each other, with a line of symmetry through *zero*.**

Absolute value is the distance a number is from zero (0) on a number line.

2. **The absolute value of 4 is *4* because it is 4 units from 0 on the number line.**



3. **The absolute value of −4 is *4* because it is 4 units from 0 on the number line.**



4. **The values 4 and −4 are opposites. Each value is 4 units from zero. A number and its opposite have the same *absolute value*.**

4.8.4 Guided Practice: Absolute Value and Zero

Absolute value is represented using the notation $|a|$. The notation $|a|$ is read, "the absolute value of a."

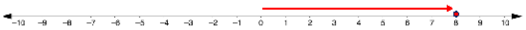
1.

a. The $|-8| = 8$ because the distance from -8 to 0 is 8.

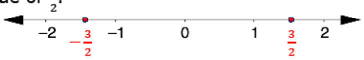
b. What other number has the same absolute value as -8?

8

c. Use the number line to show the other number that has the same absolute value as -8.



2. On the number line, plot and label all numbers with an absolute value of $\frac{3}{2}$.



3. Hector and his partner, Salih, were given the following two problems to complete.

$|5.5| = 5.5$ $\qquad$ $|-3.2| = 3.2$

3.2 is also correct

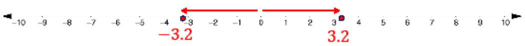
Hector and Salih had the same answer for one of the blanks but two different answers for the other. Their teacher told them they both had the correct answers.

a. Which problem can have more than one correct answer?

The absolute value that is equal to 3.2.

b. Use the number line to explain how this is possible.

3.2 and −3.2 are the same distance from 0 on the number line, so they both have an absolute value of 3.2.



4.8.4 GUIDED PRACTICE: ABSOLUTE VALUE AND ZERO

Component Overview: *Students practice finding the absolute value of numbers and plotting on a number line.*

ENRICHMENT

How do you use a number line to determine the distance a number is from zero?

4.8.5 YOUR TURN!

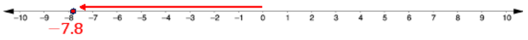
Component Overview: *Students engage in additional practice determining the absolute value and plotting those values on a number line.*

QUESTIONING

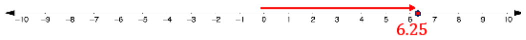
Challenge students to write a real-world example to represent the rational numbers and their absolute values from question 1. Remind students to explain what represents zero in their examples. For example, a student's home is zero, school is 3 miles north (+3), the grocery store is 5 miles south (-5).

4.8.5 Your Turn!

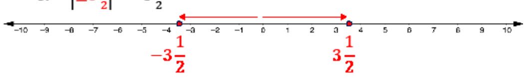
1. **Complete each statement and model it on the number line given.**
   a. $|-7.8| = 7.8$



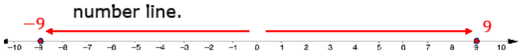
   b. $|6.25| = 6.25$



   c. $\left|\pm 3\frac{1}{2}\right| = 3\frac{1}{2}$



2. **Choose any rational number between −10 and 10.**
   a. **Plot the number and then plot its opposite on the number line.**



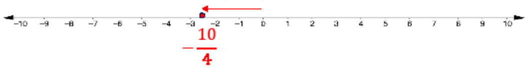
   Answers will vary.

   b. **What is the relationship between the absolute values of these rational numbers?**
   Both values are the same distance from 0 on the number line, so their absolute values are equivalent.

4.8.6 Wrap-Up: Ticket Out the Door

1. **Complete the statement and model it on the number line.**

$$\left|-\frac{10}{4}\right| = \frac{10}{4}$$



2. **What must be true about two numbers that have the same absolute value?**
   Answers will vary. Numbers with the same absolute value are opposites, the same distance from zero in opposite directions.

4.8.6 WRAP-UP: TICKET OUT THE DOOR

Component Overview: *Students demonstrate their mastery of lesson learning targets. Use data from this formative assessment to determine which students need remediation before the next lesson.*

TEACHER MOVES

Consider additional information you may want to know from students and add targeted questions to elicit those responses. For example, to elicit responses specific to 10/4 and -10/4 being the same distance from zero, use prompts like: “Write a verbal description for the location of each value on the number line. Compare their location.”